

Sample Churn Risk & Retention Report

Client-ready churn analysis package generated from the demo pipeline

XGBoost

SHAP explainability

PDF + CSV deliverables

Rows analyzed

500

Synthetic SaaS customer file

Observed churn

23.0%

Historical target rate

High-risk customers

33

Priority accounts

Top 20% capture

64.35%

Churners found in top risk bands

Executive Read

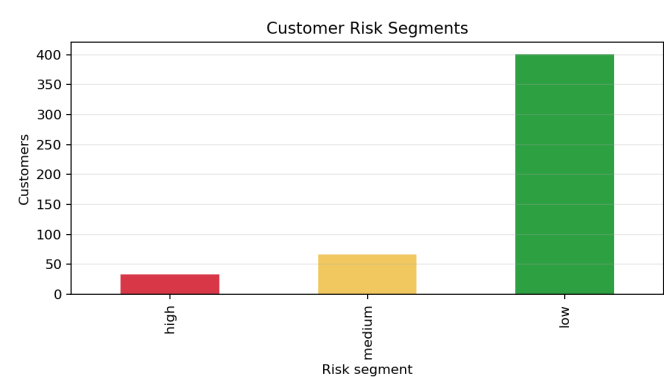
The model ranks customers by churn risk and turns the score into a retention queue. The highest-risk segment should be reviewed first because it concentrates the largest expected churn probability. This sample report is not just a model output. The delivery includes a data quality check, model metrics, lift analysis, SHAP-based driver explanations, and a customer-level action file.

Recommended next action: start with the top-risk customers, review their likely drivers, and assign outreach by priority.

Delivered files

predictions.csv
retention_recommendations.csv
risk_segments.csv
lift_table.csv
feature_importance.csv
shap_feature_importance.csv
data_quality_report.md
metrics_report.md
client_report.pdf

Risk Segment Snapshot



Segment	Customers	Avg. churn probability
High	33	77.61%
Medium	66	53.84%
Low	401	12.43%

The high-risk group is intentionally small. The point is not to label everyone as risky, but to identify the accounts that deserve immediate retention attention.

Model Proof

ROC AUC

0.77

Ranking quality

PR-AUC

0.516

Useful for imbalanced churn

Precision

0.462

High-risk predictions

F1

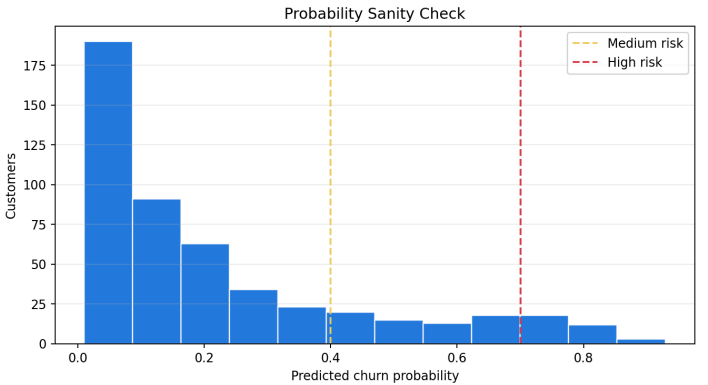
0.255

Precision/recall balance

Lift analysis

Decile	Customers	Churners	Capture
1	50	45	39.13%
2	50	29	64.35%
3	50	19	80.87%
4	50	9	88.70%
5	50	5	93.04%

The top decile contains 39.13% of all churners. The top 20% contains 64.35%. This is the business case for ranked retention work.



Probability distribution helps catch overconfident or collapsed scoring.

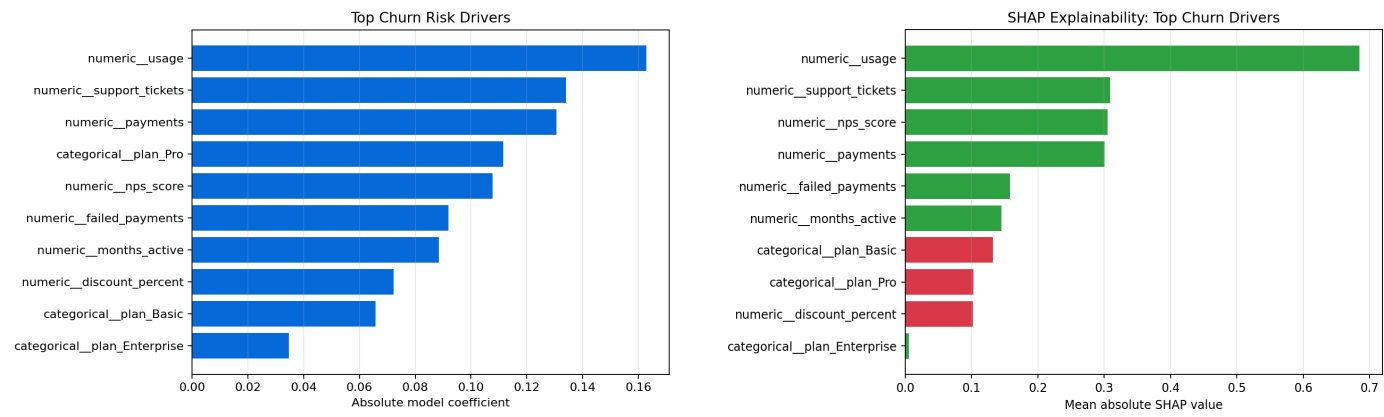
Top Recommended Accounts

Customer	Segment	Risk	Likely drivers	Priority
CUST-00335	High	92.95%	usage, support tickets, payments	1
CUST-00084	High	89.02%	usage, support tickets, payments	1
CUST-00027	High	85.32%	usage, support tickets, payments	1
CUST-00323	High	85.00%	usage, payments, plan	1
CUST-00039	High	83.31%	usage, support tickets, payments	1

How to use this queue

High-priority accounts should receive targeted intervention before renewal risk increases. The full CSV includes the recommended action text for every scored customer.

Explainability



Feature importance shows the model's strongest global signals. SHAP adds directionality so the report can explain why risk moves up or down instead of only returning a score.

Data Quality Check

Check	Result
Rows	500
Columns	10
Duplicate customer ids	0
Missing values	0
Leakage warnings	None detected

Scope note

This package is designed for a one-time churn analysis from a provided CSV. It does not include live CRM integration, scheduled retraining, or guaranteed model accuracy. Those are separate scopes.

What Makes This Delivery Different

Business-ready outputs

PDF report, ranked action queue, charts, and CSV exports.

Reproducible package

Config snapshot, model card, manifest, validation, and source code.